

the LEDs by pulling port PD6 of IC6 low through DIP1 switch.

For manual mode there are two joysticks connected at ADC channels of the controller (PC0 and PC1) at remote module. By using the joystick, robot movements can be controlled.

Move your robot towards the fire source. Pull PD5 of IC6 to low through DIP1 switch to switch on the

water pump.

For troubleshooting the circuit, check to ensure voltages at various test points are as per Table II.

Applications

1. By replacing IR photo sensors with the thermal image-processing camera, the pump with a fire extinguishing mechanism and by making the

body of the robot fire-proof, it can be used for fire extinguishing purposes. It can also be used in assisting fire fighters, thus helping them reach inaccessible places to save more lives.

2. Increasing the number of sensors will improve the resolution and accuracy of the system further.

3. In autonomous mode, it can detect a fire, if any, and extinguish it without any assistance.

4. The RF module makes it possible for the operator to control the robot manually from a distance, thus allowing surveillance facility.

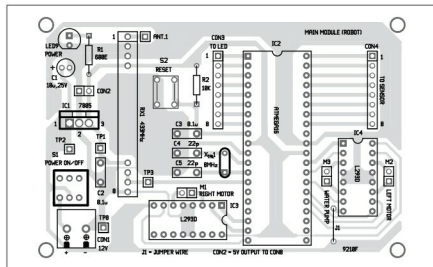


Fig. 7: Component layout of the PCB of main module (robot)

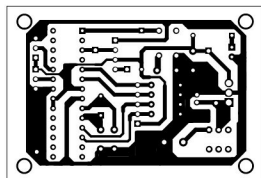


Fig. 8: Actual-size PCB of RF remote module

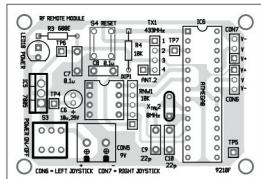


Fig. 9: Component layout of the PCB of the RF remote module

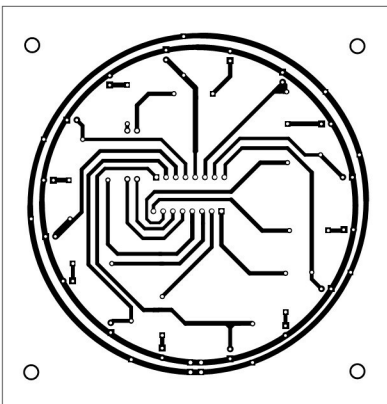


Fig. 10: Actual-size PCB of the sensor module